

Design Document: RL-Driven Physical Activity Interventions

A Configurable Simulation Framework

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Abstract

Physical activity (PA) interventions delivered via wearable devices show promise for improving public health outcomes, but personalising intervention timing and content at scale remains an open challenge. Reinforcement learning (RL) offers a principled approach to learning intervention policies that adapt to individual behaviour over time. We present **rl-health-interventions**, a configurable simulation framework that models PA interventions as a Markov Decision Process (MDP). Researchers define their dataset schema, MDP dynamics, user behaviour models, and agent configurations entirely through YAML config files—no source code changes required. The framework enables systematic comparison of RL agents (Thompson Sampling, DQN, PPO) and intervention strategies on synthetic and real-world wearable data. This document specifies the framework architecture, the MDP formalisation, and key design decisions.

1 Introduction

1.1 Background

Physical inactivity is a leading risk factor for chronic disease worldwide. Just-in-time adaptive interventions (JITAIs) delivered through wearable devices aim to provide the right support at the right moment, leveraging real-time sensor data to trigger personalised prompts. However, designing effective intervention policies—deciding *when*, *what*, and *how much* to intervene—is a sequential decision-making problem that traditional rule-based approaches handle poorly.

Reinforcement learning provides a natural framework for this problem: an agent learns a policy that maps user state (step count, sleep, heart rate, time of day) to an intervention action (motivational prompt, goal reminder, rest suggestion), optimising long-term behaviour change rather than short-term response. The key challenge is that training RL agents requires either expensive real-world trials or a realistic simulation environment.

1.2 Problem Statement

There is no standardised, configurable simulation platform for evaluating RL agents on PA intervention tasks. Existing work uses bespoke simulators tied to specific datasets or papers. This makes cross-paper comparison difficult and limits reproducibility.

1.3 Goals

This project delivers:

1. A **config-first simulation framework** where the dataset schema, MDP, user behaviour, and agents are specified entirely via YAML config files.

2. A **configurable MDP environment** supporting multi-timescale rewards (daily steps + delayed clinical measures).
3. A **rule-based user simulator** grounded in behavioural theory, with configurable archetypes and fatigue mechanisms.
4. A **library of RL agents** sharing a common interface, from simple baselines (Thompson Sampling) to deep RL (DQN, PPO).
5. An **experiment runner** for reproducible multi-trial, multi-agent comparisons.

1.4 Scope

Phase 1 (8 weeks) delivers a working CLI: `uv run rl-health-interventions --config experiment.yml` that trains agents and produces a results table. Phase 2 (stretch) adds LLM-based user simulation and validation against real data.

1.5 Literature Gaps

A systematic literature review identified five gaps this framework addresses:

1. No LLM simulation validated for wearable health data
2. No delayed-feedback bandit for 3-week clinical measures
3. No non-stationary adaptation for PA interventions
4. No standardised MDP/benchmark for PA JITAI RL
5. No offline RL for PA interventions with real wearable data

These gaps are *experiments run on the framework*, not part of the framework itself.

2 Architecture

2.1 Design Principles

- **Config-first:** Every component reads from YAML. New dataset or intervention design = new config, not new code.
- **Pluggable components:** Each family (transitions, rewards, agents, response models) uses an ABC + registry pattern. Adding a new component requires one file, one registry entry, and a config string reference.
- **Minimal dependencies:** Python 3.11, uv, Polars, Pydantic, NumPy. No Stable Baselines or Gymnasium unless proven necessary.
- **Reproducibility:** Config snapshots, fixed seeds, deterministic evaluation.

2.2 Component Overview

The framework is organised into five subphases, each delivering a pluggable component:

1. **Data Layer (1A):** Ingests CSV/Parquet files via Polars lazy scans, applies config-driven feature engineering, and materialises a numpy-backed `Dataset`. Provides a `StateView` bridge for the simulation loop.
2. **MDP Environment (1B):** Exposes a clean `step(state, action) → (next_state, reward, done)` API. Pluggable `TransitionModel` and `RewardHandler` components are selected from a registry by config string.
3. **User Simulation (1C):** Rule-based behavioural archetypes (goal-driven, social responder, resistant, stable maintainer) with backlash and fatigue mechanisms. A `ResponseModel` maps (state, action, profile) to a next state.

4. **Agent Library (1D):** Common Agent ABC with `select_action(state) → int` and `update(...)`. Thompson Sampling is the MVP gate; deep RL agents (DQN, PPO, CQL, IQL) are stretch.
5. **Experiment Runner (1E):** Wires all components via a single `ExperimentFactory.build(config)`. Runs multi-trial, multi-agent experiments with config snapshot reproducibility.

2.3 Dependency Graph

Tracks A (1A) and B (1B) run in parallel. Dataset exploration runs alongside both. After 1A and 1B complete, tracks merge at 1C (user simulation) and 1D (agent library). The final track is 1E (experiment runner), which depends on 1C and 1D.

2.4 Validation

The factory applies three layers of validation before the main experiment loop: (1) Pydantic schema parsing rejects malformed configs at load time; (2) component compatibility checks verify registry keys and cross-references; (3) a single dummy `step()` catches wiring errors and shape mismatches.

2.5 Test Organisation

Tests mirror the component module structure (`tests/unit/transitions/`, `tests/unit/rewards/`, etc.), not subphase numbers. Every ABC method has at least one test. Integration and end-to-end tests cover factory wiring and the full experiment pipeline.

3 MDP Formalisation

This section specifies the Markov Decision Process (MDP) for the first iteration of the simulation framework. The MDP is defined as the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

3.1 State Space $\mathcal{S}$

The state represents the current context of a user at decision epoch t . States are constructed from wearable-device signals, user profile information, and intervention history:

$$s_t = \{\mathbf{x}_t^{\text{wearable}}, \mathbf{x}_t^{\text{context}}, \mathbf{x}_t^{\text{history}}, \mathbf{x}^{\text{profile}}\}$$

Variables are normalised to $[0, 1]$ before being passed to the agent. The body measure is only observed every 3 weeks (sparse reward signal).

3.2 Action Space $\mathcal{A}$

$$\mathcal{A} = \{a_0, a_1, a_2, a_3, a_4, a_5\}$$

Exactly one action per decision epoch. The “no message” action is critical for managing notification burden.

3.3 Transition Function P

The transition function models the probability over next states:

$$P(s_{t+1} \mid s_t, a_t) : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$$

In Phase 1 (rule-based), transitions follow a behavioural response model:

Table 1: State variables

Variable	Type	Description
steps_t	$\mathbb{R}_{\geq 0}$	Step count, past 24h
hr_t	$\mathbb{R}_{\geq 0}$	Heart rate (or resting)
sleep_hours_t	$\mathbb{R}_{\geq 0}$	Previous night sleep
sedentary_min_t	$\mathbb{R}_{\geq 0}$	Sedentary minutes today
time_of_day_t	$\{0, \dots, 23\}$	Current hour
day_of_week_t	$\{0, \dots, 6\}$	Day of week
goal_progress_t	$[0, 1]$	Fraction of daily goal met
burden_t	$\mathbb{Z}_{\geq 0}$	Interventions sent today
a_{t-1}	$\mathcal{A}$	Previous action
response_{t-1}	$\{0, 1\}$	Previous response
body_measure_k	$\mathbb{R}$	Clinical measure (q3wks)
age	$\mathbb{Z}_{>0}$	Static profile
gender	$\{0, 1\}$	Static profile
baseline_activity	$\{0, 1, 2\}$	Low / medium / high

Table 2: Action space

Action	Label	Description
a_0	No message	Do not intervene
a_1	Motivational prompt	Generic encouragement
a_2	Walking suggestion	Recommend a short walk
a_3	Goal reminder	Remind of step goal
a_4	Recovery suggestion	Stretch or rest
a_5	Progress feedback	Report goal progress

$$\Delta \text{steps}_{t+1} = f(s_t, a_t, \theta_{\text{user}})$$

where θ_{user} parameterises one of four behavioural archetypes:

- **Goal-driven:** Responds to reminders and feedback. Proportional to goal progress.
- **Social responder:** Responds to motivational prompts. Models social desirability bias.
- **Resistant:** Low response. Fatigue accumulates quickly.
- **Stable maintainer:** Already active. Low marginal gain.

Burden evolves as:

$$\text{burden}_{t+1} = \begin{cases} \text{burden}_t + 1 & \text{if } a_t \neq a_0 \\ 0 & \text{if } a_t = a_0 \end{cases}$$

When $\text{burden} > B_{\text{max}}$, response probability decays linearly.

3.4 Reward Function R

3.4.1 Immediate Reward

$$R_t^{\text{immediate}} = \alpha \cdot \Delta \text{steps}_t - \beta \cdot \mathbf{1}_{\{a_t \neq a_0\}} + \lambda \cdot \text{goal_progress}_t$$

3.4.2 Delayed Reward (Body Measures)

Clinical body measures arrive every 3 weeks ($t = 3k$):

$$R_{3k}^{\text{delayed}} = \eta \cdot \text{BM.improvement}_{3k}$$

Total reward:

$$R_t = \begin{cases} R_t^{\text{immediate}} + R_t^{\text{delayed}} & t \equiv 0 \pmod{3} \\ R_t^{\text{immediate}} & \text{otherwise} \end{cases}$$

The compound reward (immediate + delayed) is a central research question. Weights $(\alpha, \beta, \lambda, \eta)$ are configurable experiment parameters.

3.5 Discount Factor γ

$$\gamma \in [0.9, 0.95]$$

A high discount factor encourages long-term behaviour change over short-term step increases. Appropriate for health habit formation.

3.6 Decision Epoch

Daily decisions. This aligns with the weekly intervention cadence, the 3-week body measure delay (3 epochs), and practical deployment constraints.

4 Configurability

All MDP components are exposed in the config schema: state variable selection, action subsets, reward weights, transition archetypes, fatigue threshold, discount factor, and epoch frequency. New datasets or intervention designs require a new config file, not new code.

Table 3: Configurable parameters

Category	Parameter	Default
State	Variable selection	All
Actions	Available subsets	$\{a_0, \dots, a_5\}$
Reward	$(\alpha, \beta, \lambda, \eta)$	$(1.0, 0.1, 0.5, 1.0)$
Transition	Archetype	goal-driven
Fatigue	$B_{\max}$	3
Discount	γ	0.95
Epoch	Frequency	Daily

5 Decision Log

Key decisions made during the design phase:

5.1 Open Decisions

1. Installable package or repo-based distribution?
2. Does the proposed MDP look like a reasonable starting point?
3. Which datasets to target first for real-data validation (Phase 2)?

Table 4: Design decisions

#	Decision	Status
1	Config format: YAML	Confirmed
2	Interface: CLI + config files; web UI = stretch	Confirmed
3	Python 3.11 + uv for package management	Confirmed
4	Synthetic data for Phase 1 (real data requires 4-8 week applications)	Confirmed
5	Thompson Sampling as MVP agent	Confirmed
6	Daily decision epochs	Confirmed
7	Multi-timescale reward: immediate (daily steps) + delayed (3-week body measures)	Confirmed
8	Four behavioural archetypes: goal-driven, social responder, resistant, stable maintainer	Confirmed
9	Pluggable component architecture (ABC + registry pattern)	Confirmed

6 Data Sources

6.1 Phase 1: Synthetic Data

All Phase 1 experiments use synthetic data generators parameterised from published statistics. This avoids the 4-8 week lead time for real dataset access.

6.2 Phase 2: Real Data Targets

Two public datasets are under consideration:

1. **All of Us Fitbit Dataset** (Nature Medicine 2026): 59,000+ participants, Fitbit data, 14-year span. Access via cloud workbench only.
2. **UK Biobank Accelerometer Dataset** (npj Digital Medicine 2024): 700,000+ person-days, wrist-worn tri-axial accelerometer. Requires institutional application (4-8 weeks).

Supplementary: HeartSteps V1/V2 micro-randomized trials provide intervention response data for calibrating the user simulator.

7 Timeline

Table 5: Phase 1 milestones

Week	Milestone
1–2	Data layer (1A) + MDP environment (1B)
3–4	User simulation (1C) + Agent library (1D)
5–6	Experiment runner (1E) + end-to-end tests
7–8	Documentation, examples, handover

8 Future Work

- LLM-based user simulation (Phase 2 stretch)
- Offline RL agents (CQL, IQL) for batch policy learning
- Non-stationary environment adaptation
- Cross-dataset transfer experiments
- Integration with real-world JITAIs